

Multiple Choice (10 marks)

1. A combinational PLD with a fixed AND array and a programmable OR array is called a
 - (a) ROM
 - (b) FPGA
 - (c) PLA
 - (d) PLD
 - (e) EEPROM
2. Evaluate $f(t)$ if $F(s) = [(3s^2 + 17s + 47) / \{(s + 2)(s^2 + 4s + 29)\}]$
 - (a) $f(t) = [e^{-2t} + 2.24e^{-2t}\sin(5t - 26.6^\circ)]u(t)$
 - (a) $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$
 - (a) $f(t) = [e^{-2t} - 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$
 - (a) $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t + 26.6^\circ)]u(t)$
3. In a dc machine 6 pole wave winding is used. The number of parallel paths are?
 - (a) 2
 - (b) 8
 - (c) 7
 - (d) 12
 - (e) 3
4. The directivity of an antenna is 50 and the antenna operates at a wave length of 4 meters. What is its maximum effective aperture ?
 - (a) 4 meter^2
 - (b) 100 meter^2
 - (c) 50 meter^2
 - (d) 40 meter^2
 - (e) 16 meter^2
5. Snubber circuit is used with SCR
 - (a) either series or parallel.

- (b) in series.
- (c) in an alternating manner.
- (d) in series-parallel.
- (e) in anti series.

True / False (10 marks)

Answer True or False

6. True or False: The product AB equals $|96| |55| |138|$ when calculated using the given absolute values.
7. True or False: A 7-hp motor operating at 6.3 kilowatts typically achieves an efficiency of 90%.
8. True or False: $V_{ab}(t)$ can be expressed as $V_{ab}(t) = 30 te^{\dagger - 4t} + 30e^{\dagger - 4t} - 36e^{\dagger - 2t}$.
8. True or False: The number of rings in the Bohr model of any element is determined by the number of isotopes of that element.
9. True or False: The inverse Laplace transform of $1 / \{s(s^2 + 1)\}$ is $t - \sin t$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the mass of air in a closed chamber with dimensions $35 \text{ ft} \times 20 \text{ ft} \times 10 \text{ ft}$ at a pressure of 17 lb/in^2 and a temperature of 75°F , assuming ideal gas behavior. Explain your methodology and reasoning.
11. A 50-hp, 500-volt shunt motor operates at no load with a line current of 4.5 A. Calculate the horsepower output and efficiency at full load, considering the given resistances and brush drop. Discuss your methodology and reasoning.

End of Paper